SMART INDIA HACKATHON 2025

- **Problem Statement ID – SIH25029**
- **Problem Statement Title - Authenticity validator for Academia**
- **Theme - Smart Education**
- **PS Category- Software**
- **Team ID -**
- **Team Name - Lumora**

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CertVerify

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PROPOSED SOLUTION

Student

- → Upload Document
- → Receive a unique verification token as proof.

Recruiter

- → DigiLocker (API verification).
- → Token verification (already verified)
- → Doc upload & verification pipeline

Verification pipeline

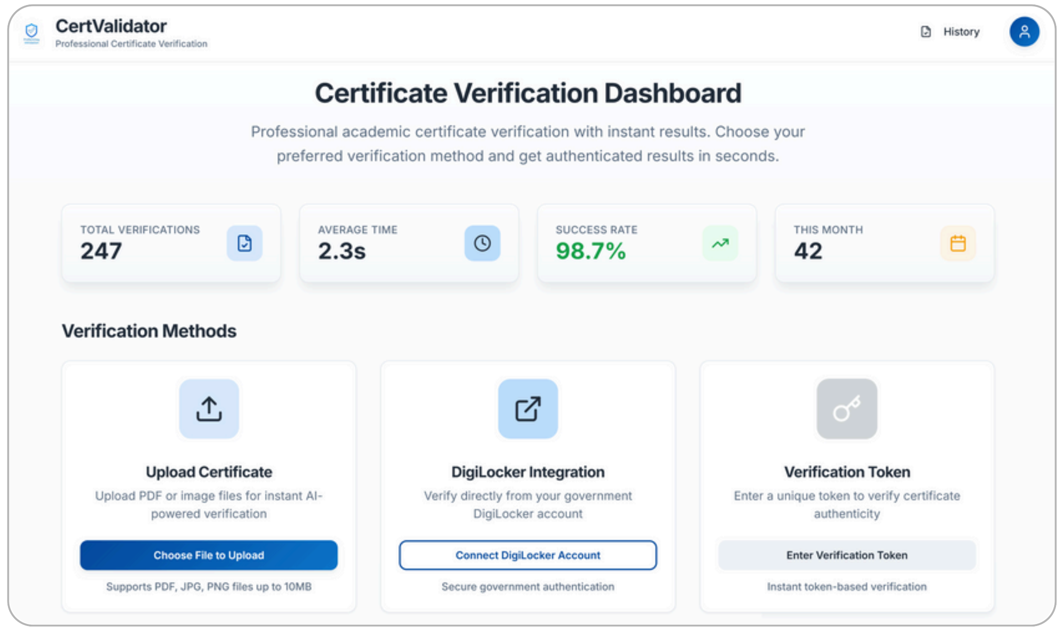
1. OCR + Database matching.
2. Blockchain-based validation
3. AI/ML forgery detection

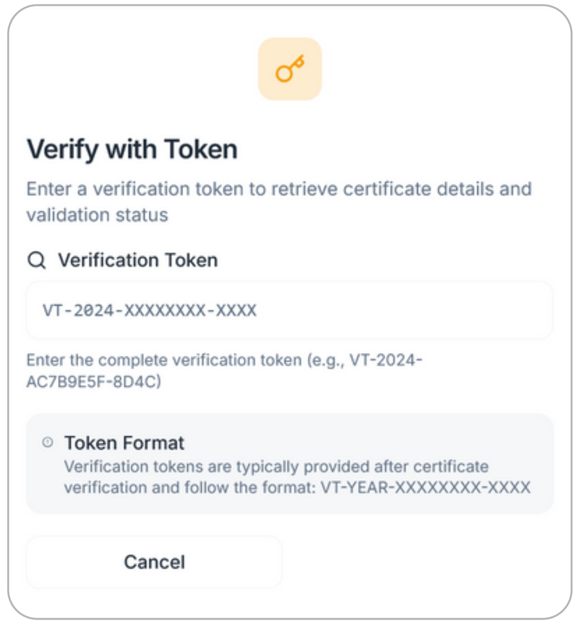
ADRESSING PROBLEM

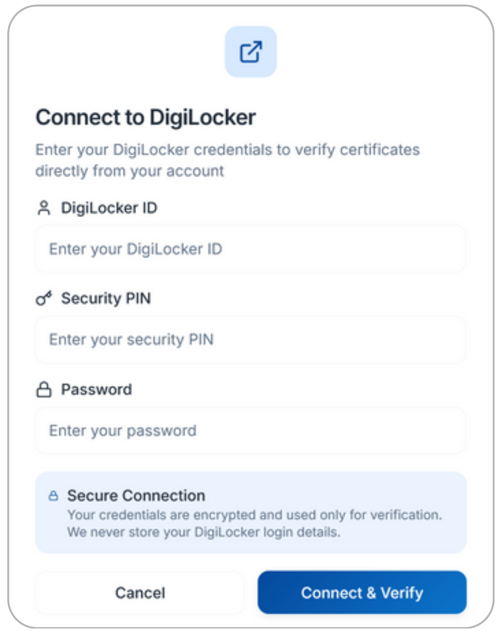
- Stops fake/forged certificates with **multi-step checks.**
- Gives recruiters quick and reliable verification.
- **Cuts delays** of manual inspections.
- Ensures **uniform verification** across institutions.
- **Recruiter verified** via GST/CIN, domain,cross-platform
- **Safeguards** academic integrity.

INNOVATION & UNIQUENESS

- Combines **3 Step** Verification for all **edge cases.**
- **Fallback paths :** manual review if OCR/DB fails.
- Token system for **reusable verification**.
- Supports both **old** and **new** certificates.
- **Scalable** for state/nationwide rollout.







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TECHNICAL APPROACH

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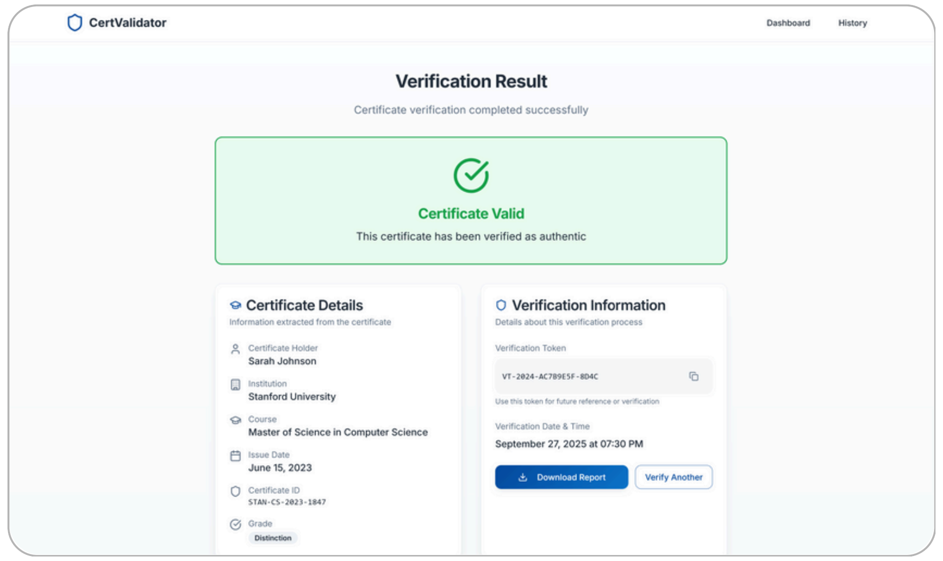
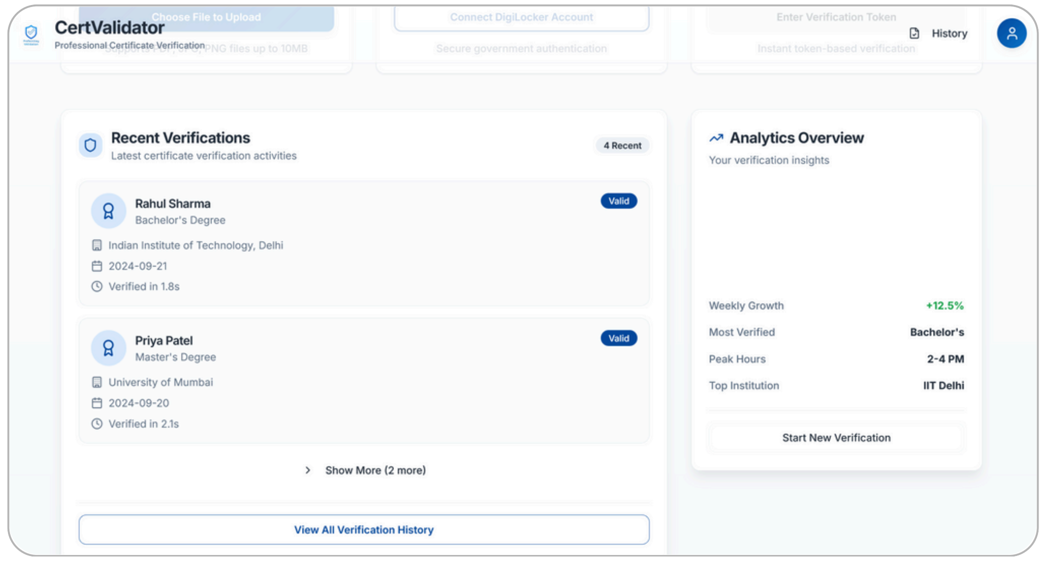
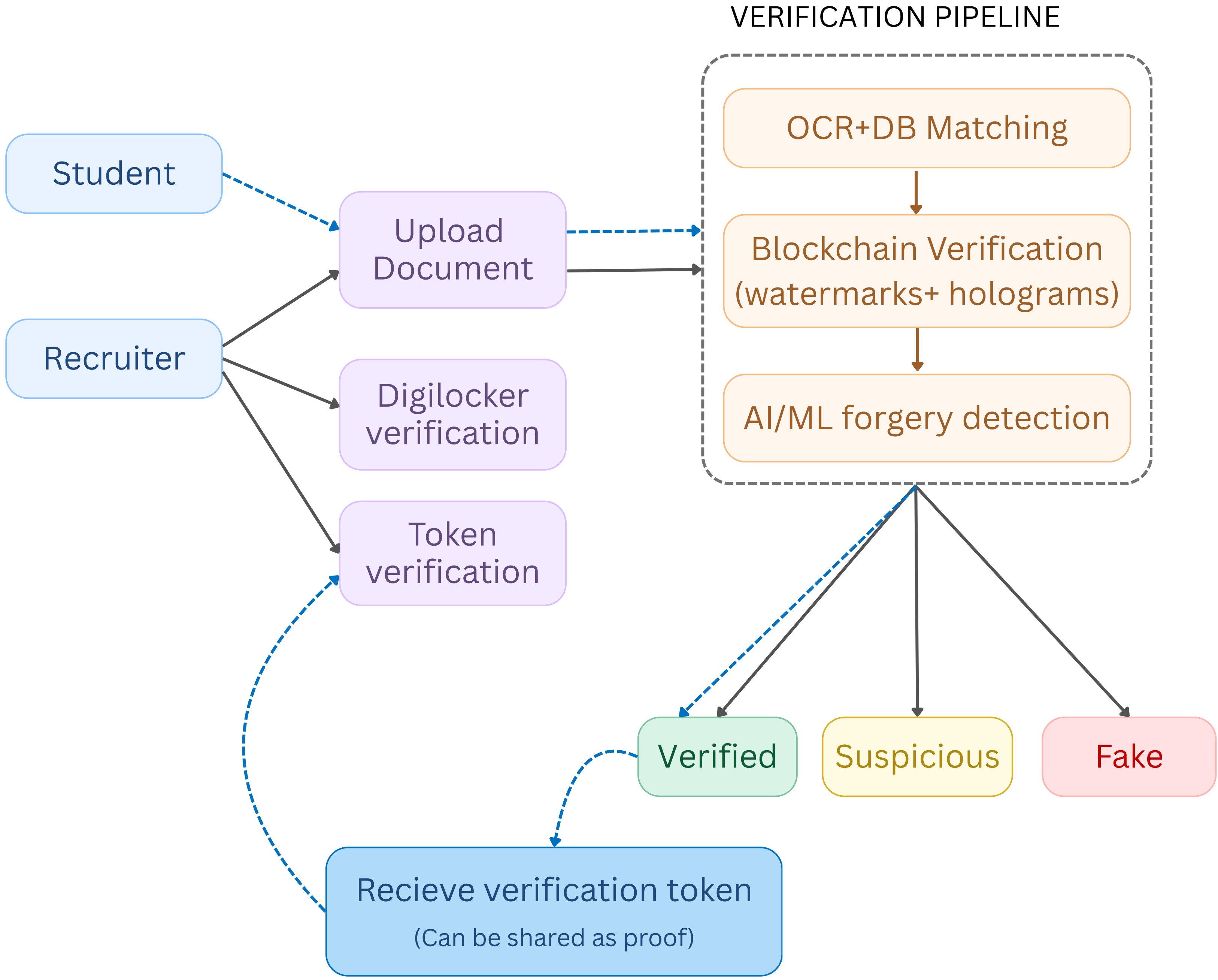
TECHNOLOGIES

Frontend	React.js
Backend	Python (Django / FastAPI)
OCR	Tesseract OCR / Google Vision API
Blockchain	Hyperledger / Ethereum smart contracts
AI/ML Models	CNNs+Transformers for forgery & anomaly detection
Database	MySQL / PostgreSQL for structured data
Integration	DigiLocker API for government-verified credentials





FLOWCHART



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FEASIBILITY AND VIABILITY

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FEASIBILITY

- **Technical**: OCR, AI/ML, and Blockchain are proven tools for secure verification
- **Operational**: APIs and dashboards enable easy rollout for both digital & old records.
- **Security**: Encrypted, access-controlled, and compliant with IT laws & GDPR.
- **Scalability**: Cloud + private blockchain keeps costs low and supports nationwide use

CHALLENGES & RISKS

- **OCR Errors** : Low-quality scans may cause text misreading.
- **Legacy Systems** : Many universities still use old/manual records.
- **Blockchain Cost** : Public blockchain can be slow and expensive.
- **Data Privacy**: Student data is sensitive and must be well-protected.

MITIGATION STRATEGIES

- **Fallback** : Manual review for poor scans or OCR errors.
- **Cost Control** : Use private blockchain (e.g., Hyperledger) to cut costs.
- **Security** : AES-256 encryption + access control + regular audits
- **Adoption** : Start with pilot universities, then expand gradually.

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IMPACT AND BENEFITS

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IMPACT

- **Institutions (Colleges/Universities) →** Protect academic reputation, reduce fake certificate misuse.
- **Recruiters & Employers →** Faster and more reliable candidate verification.
- **Students →** Genuine achievements recognized fairly, no competition with fake degree holders.
- **Government & Agencies →** Easier fraud tracking, supports Digital India & education reforms.

BENEFITS

- **Trust & Transparency →** Builds confidence in educational and hiring ecosystem.
- **Time & Cost Savings →** Eliminates delays of manual checks and reduces administrative costs.
- **Fraud Prevention →** Detects and blocks forged/tampered certificates early.
- **Fairness & Integrity →** Creates equal opportunities for deserving candidates.

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RESEARCH AND REFERENCES

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CASE STUDIES

- Large-Scale Fraud in Uttar Pradesh: In July 2025, the Supreme Court flagged large-scale fake certificate fraud in Uttar Pradesh, with reports indicating that 8,000 certificates from Agra University were found to be fake. *Source : TOI*
- 17 teachers in Jharkhand were dismissed for using invalid certificates to secure their positions. The authorities took strict action against these individuals to maintain the integrity of the education system *Source : TOI*

RESEARCH PAPER

- Shivam Gangwar, Anushka chaurasia, "Blockchain-based Authentication and Verification System for Academic Certificate using QR Code and Decentralized Applications." International Journal of Computer Applications (0975- 8887) Volume 186- No.26, June 2024.
- Rane, Mukul & Singh, Shubham & Singh, Rohan & Amarsinh, Vidhate. (2020). Integrity and Authenticity of Academic Documents Using Blockchain Approach. ITM Web of Conferences. 32. 03038. 10.1051/itmconf/20203203038.
- Singh, Rohan. "Integrity and Authenticity of Academic Documents Using Blockchain Approach." ITM Web of Conferences, EDP Sciences.

Nearly 28% of job applicants submitted fake degrees, highlighting widespread academic fraud (2020 report).